Revolutionary Rotary Machine Technology

A breakthrough rotary pumping mechanism delivering unprecedented efficiency as both compressor and internal combustion engine

> ***Compressor: Working Prototype | Engine: Design Phase***

Core Innovation

A novel rotary pumping mechanism that eliminates the limitations of traditional piston and rotary designs.

Key Numbers

Metric	Value
Energy Savings	**30%**
Weight Reduction	**2-3x lighter**
Compressor Electricity Cost Share	**70-80%***
Annual Savings per 100HP Unit	**$14K+**

**Industry average according to U.S. Department of Energy studies*

Rotary Machine Concept

Proprietary Design - Patent Pending

Simplified Architecture

60%+ fewer components than piston engines. Results in **20%+ longer lifespan** and dramatically reduced maintenance costs.

Double Power Cycles

Two-stroke equivalent delivers **2x power output** per rotation. Achieves **104 kW/L** vs 40-60 kW/L for piston engines.

30% Lower Energy Use

Reduced friction losses translate to **30% electricity savings**. For industrial compressors, this means **$14,000+/year** in reduced operating costs.

Compressor Application

Working prototype demonstrating revolutionary performance improvements.

Metric	Value
Electricity Savings	**30%**
Weight Reduction	**60%+**
Quieter Operation	**4x**
Lower Friction	**30%+**
Increased Durability	**20%+**

Massive Cost Reduction Potential

Electricity accounts for **70-80% of a compressor's lifetime operating cost**. Our 30% energy savings directly translates to dramatic cost reductions.

$14,400+/year

Potential savings per 100 HP industrial compressor unit

10-Year Savings per Unit	Typical Annual Energy Cost*	ROI Timeline
$144,000+	**~$48,000**	**Months, Not Years**

**Based on 100HP compressor operating at $0.10/kWh, 8,760 hours/year*

Industry Applications

- **Military Vehicles** - HVAC systems and pneumatic equipment for armored vehicles requiring compact, reliable air supply
- **Aviation** - Pilot suit air supply systems where weight and reliability are critical factors
- **Naval Applications** - Diving air systems and compact installations for space-constrained marine environments
- **Rescue Equipment** - Portable pneumatic tools for emergency response requiring lightweight, reliable operation

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Engine Application

Projected specifications based on design simulations for 1L displacement

Engine Cross Section

Internal Architecture

Engine Comparison

Specification	Traditional Piston	Rotary Machine
Mass per Liter	65-80 kg/L	**30 kg/L**
Power Output	40-60 kW/L	**104 kW/L**
Max RPM	6,000-8,000	**10,000+**
Noise Level	85-95 dB	**<75 dB**
Vibration	High	**Minimal**

Target Industries

- **Defense Drones** - Combat and reconnaissance UAVs requiring high power-to-weight ratio and extended range
- **Armored Vehicles** - Tanks and APCs benefiting from compact, powerful engines with reduced thermal signature
- **Marine Vessels** - Cutters and patrol boats requiring quiet, efficient propulsion systems
- **Hybrid Powertrains** - Range extenders for electric vehicles combining efficiency with compact packaging

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Technical Specifications

Detailed comparison with conventional piston technology.

Parameter	Piston Engine	Rotary Machine	Advantage
Chamber Diameter (1L)	~80mm bore	200mm	Simplified manufacturing

Parameter	Piston Engine	Rotary Machine	Advantage
Displacement	4x250cc cylinders	4x250cc chambers	Equivalent capacity
Engine Mass	65-80 kg/L	**30 kg/L**	**2-3x lighter**
Power Density	40-60 kW/L	**104 kW/L**	**2x more powerful**
Operating Noise	85-95 dB	**<75 dB**	**2-4x quieter**
Maximum RPM	6,000-8,000	**10,000+**	**30-50% higher**
Friction Losses	Baseline	**-30%**	**Higher efficiency**
Component Count	High (valvetrain, etc.)	**Minimal**	**Better reliability**

Benefits Summary

Comprehensive advantages across economic, environmental, and operational dimensions.

Economic Benefits

- **30% electricity savings** - the biggest operating cost
- $14,000+/year savings per 100HP compressor unit
- Lower production costs due to simpler design
- Reduced maintenance requirements
- Extended service intervals
- Fewer spare parts needed

Environmental Benefits

- **30% less energy consumption** = smaller carbon footprint
- Limited or eliminated oil usage
- Higher fuel efficiency
- Reduced emissions potential
- Smaller material footprint (60% lighter)

Operational Benefits

- Significantly quieter operation
- Minimal vibration
- Higher reliability
- Easy scalability
- Extended operational lifespan

Development Roadmap

Phase	Status	Description
Concept Design	Completed	Initial concept and feasibility study completed
Compressor Prototype	Completed	Working prototype demonstrating core technology
Engine Design	**In Progress**	Detailed engineering and simulation phase
Engine Prototype	Planned	Build and test first engine prototype
Commercialization	Planned	Production partnerships and market entry

Contact

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